



!!ABDUCTIVE REASONING

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The Mystery of Station Polaris V

Background Narrative:

You are an investigator reviewing an incident report from "Polaris V," a tiny, isolated Arctic research station crewed by only three specialists. The station is completely cut off from the outside world for the winter. The incident concerns the disappearance of a one-of-a-kind biological specimen: a plant genetically engineered by the lead botanist, Dr. Aris Thorne. The plant, named *Cryoflora lumina*, is unique because it emits a constant, bright blue light and generates a small but steady amount of heat, allowing it to grow directly on ice. It is priceless and the key to Dr. Thorne's career.

One morning, Dr. Thorne entered his lab to find the plant's specialized, climate-controlled containment chamber empty. The plant was gone. There are no signs of a break-in to the station itself, and the logs show no one has entered or exited for weeks. The plant must still be somewhere within the station's three connected modules: the Lab, the Habitation Module, and the Engineering Bay.

The Three Crew Members:

- **Dr. Aris Thorne (Botanist):** The plant's creator. He is brilliant but known to be emotionally volatile and under immense pressure from his funding agency to produce results.
- **Lena Petrova (Geologist):** A quiet and methodical scientist responsible for monitoring seismic activity. Her seismograph is so sensitive it can detect the slightest vibrations within the station.
- **Ben Carter (Engineer):** The station's technician. He is responsible for maintaining all systems, including the power generator, heating, and electronics. He is a known tinkerer who often works on personal projects in his spare time.

The Evidence (Your Observations):

1. The containment chamber's electronic lock was not forced. The access log shows the door was opened precisely at **03:15 AM** using a valid keycard. All three crew members have a keycard with access.

2. Dr. Thorne has been complaining for weeks that the station's main generator is unreliable, causing brief, intermittent power flickers that threaten his delicate experiments. He has formally logged multiple complaints with Ben.
 3. On the floor of the lab, a few feet from the empty chamber, is a small, crystallized patch of what appears to be **spilled salt**.
 4. Lena Petrova's seismic report for the night shows a single, anomalous event: a faint, high-frequency **buzzing vibration** that lasted for exactly 60 seconds, starting at **03:15 AM**. The vibration was localized to the Engineering Bay.
 5. Ben Carter's workstation in the Engineering Bay is unusually tidy, except for a discarded coil of copper wire and a textbook left open to a chapter on **thermoelectric generators**.
 6. The station's internal temperature logs show that the temperature in the Habitation Module briefly **dropped by 5 degrees** at **03:17 AM** before returning to normal a few minutes later. This coincided with a power draw spike from the module's heating unit.
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Your Task

Based on all the available evidence, construct the **most likely explanation** for what happened to the *Cryoflora lumina*.

In your answer, you must:

1. Identify the person you believe is responsible.
2. Explain their probable motive.
3. Connect at least **four** of the specific pieces of evidence to your conclusion, explaining how they fit together to tell a single, coherent story.

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